

Network and Data Communications

BCA — Semester 4 — — Answer Sheet

Subject Code:

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

1. What are the features of Distance Vector Routing protocol? Explain Distance Vector Routing protocol with relevant example. What are its disadvantages?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Explain the concept of Binary Phase Shift Keying. Represent bit sequence 111000110 by the following waveform. NRZ-lb. Differential Manchester

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Explain the design of Selective Repeat ARQ. Illustrate it with suitable flow diagram example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. List different categories of networks and explain any two.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What is SNMP and how does it work?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What is MAC-address? The message sequence is 1011011 and generator polynomial $G(X) = x^3 + x^2 + 1$. Calculate the transmitted encoded frame.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. How IGMP allows devices to join a multicast group? Explain.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Differentiate between reliable and unreliable protocol. Provide example of each. How can these protocols be used?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Explain different phases of packet switching. How packet switching works?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Explain Frequency Division Multiplexing with required figure.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. A pure ALOHA network transmits 200-bit frames using a shared channel with a 200-kbps bandwidth. Find the throughput if the system considering all stations together produces 1000 frames per second. What do you mean by vulnerable time of pure ALOHA?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short notes on a. Shannon Capacity b. Connectionless service

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.